Evapotranspiration signatures:

CABLE and JULES vs flux towers at 22 Australian sites

Jamie (CCRC placement), October 2026. Signatures from Table 2 (p. 6) of Gardiya Weligamage et al. (2025, EGUsphere, doi:10.5194/egusphere-2025-3373), computed with et_signatures.py on the harmonised PLUMBER2 site files. AU-Ync is excluded (its tower files are a copy of AU-ASM).

Key findings

- Both models get the big picture right. Across sites, annual ET ranges from about 260 to 1,390 mm/yr and both models track it (cross-site correlation r = 0.81 for CABLE, 0.86 for JULES). CABLE's typical annual ET is -10% of the tower's value off, JULES's +3%.
- Their errors are shared. CABLE's and JULES's annual-ET errors rise and fall together across sites (Spearman rho = 0.80). At 13 of 22 sites both models are off in the same direction. This points to shared model structure, or to the forcing and tower data, more than to one model's quirks.
- The wettest sites are missed most. At the two lowland tropical rainforests (AU-Cow, AU-Ctr) both models evaporate 26-35% less than the tower, and 12-17% less at upland AU-Rob. At AU-Ctr the tower measures 1,390 mm/yr against 948 (CABLE) and 984 (JULES).
- Tropical savannas collapse in the dry season. At AU-How, AU-DaS and AU-Dry the real savanna's ET in its three driest months is 41-53% of its wettest-month ET; CABLE's is 16-26%. This dry-season collapse also exaggerates CABLE's seasonal swing (CV_monthly +0.21 in this group). JULES does better but has the same tendency.
- Arid sites: totals right, dynamics wrong. Both models match arid-site annual ET (it is capped by rainfall), but both respond far more strongly to individual rain events than the tower does (e.g. R = 0.47/0.49 vs 0.10 at AU-Cpr). The excess responsiveness grows with dryness (rho = 0.54, p = 0.01 for CABLE).
- A clear missing process at AU-Gin. Gingin (Banksia woodland on deep sand) is the only site that evaporates more in its dry season than its wet season (ratio 1.27, peak in November), presumably from shallow groundwater. Neither model can do this: ratios 0.52 and 0.61, with peaks 2-3 months early.
- JULES is better in some respects, not all. JULES is closer on water stress (15 of 22 sites, Wilcoxon p = 0.028) and on dry-season persistence (17 of 22, p = 0.006). For annual totals it wins 14 of 22 (p = 0.14, not significant), and there is no difference for CV_annual, CV_monthly or R. JULES gets the peak month exactly right at 14 sites against CABLE's 9; CABLE tends to peak early.
- Which tower data you compare against matters. JULES's apparent over-estimates sit at sites where the tower misses most energy (rho = 0.51, p = 0.015). Against energy-balance-corrected tower ET, both models under-estimate (median -26% CABLE, -18% JULES).

1. Data and method

Data. One harmonised file per site (data/harmonised/<SITE>.nc) holds the tower observations and the CABLE and JULES PLUMBER2 runs on the same half-hourly axis. Both models were run 'out of the box' with the tower's weather as forcing.

ET. Latent heat flux is converted to evapotranspiration with a temperature-dependent latent heat of vaporisation, lambda = (2.501 - 0.00237 T) x 10^6 J/kg, summed to days, months and years. As in the paper, the tower ET is gap-filled and negative values are excluded (here: negative daily totals). A month or year counts only if 90% of its days are present.

PET. The paper uses Morton wet-environment PET from SILO. We use a substitute: Priestley-Taylor PET (alpha = 1.26) from the tower's net radiation and air temperature, the same PET for the tower and both models. This affects metric 6 (WS) and the aridity index used to order sites.

Metrics. From Table 2 we use 1 (long-term median annual ET), 2 (CV of annual ET), 4 (TSP, month of peak ET), 5 (CV of monthly ET), 6 (WS = (PET - AET)/PET) and 8 (R, event responsiveness: correlation between rain on days with more than 10 mm and the maximum daily ET in the following 0-10 days). Metrics 3 and 7 were left out on purpose. One extra diagnostic was added for this report: the dry/wet-season ratio = mean monthly ET in the 3 calendar months with least rain divided by the 3 with most. It measures whether a site keeps evaporating through its dry season.

Comparison. For each metric we compare model and tower across the 22 sites (correlation, median error, RMSE), count where each model is closer, and test the difference in absolute error with a paired Wilcoxon signed-rank test. With only 22 sites, p-values are indicative, not definitive. TSP is compared as a circular month offset (December is one month from January). CV_annual uses only the 17 sites with at least 3 complete years; the short sites are AU-Emr, AU-Lit, AU-Otw, AU-Whr, AU-Wrr.

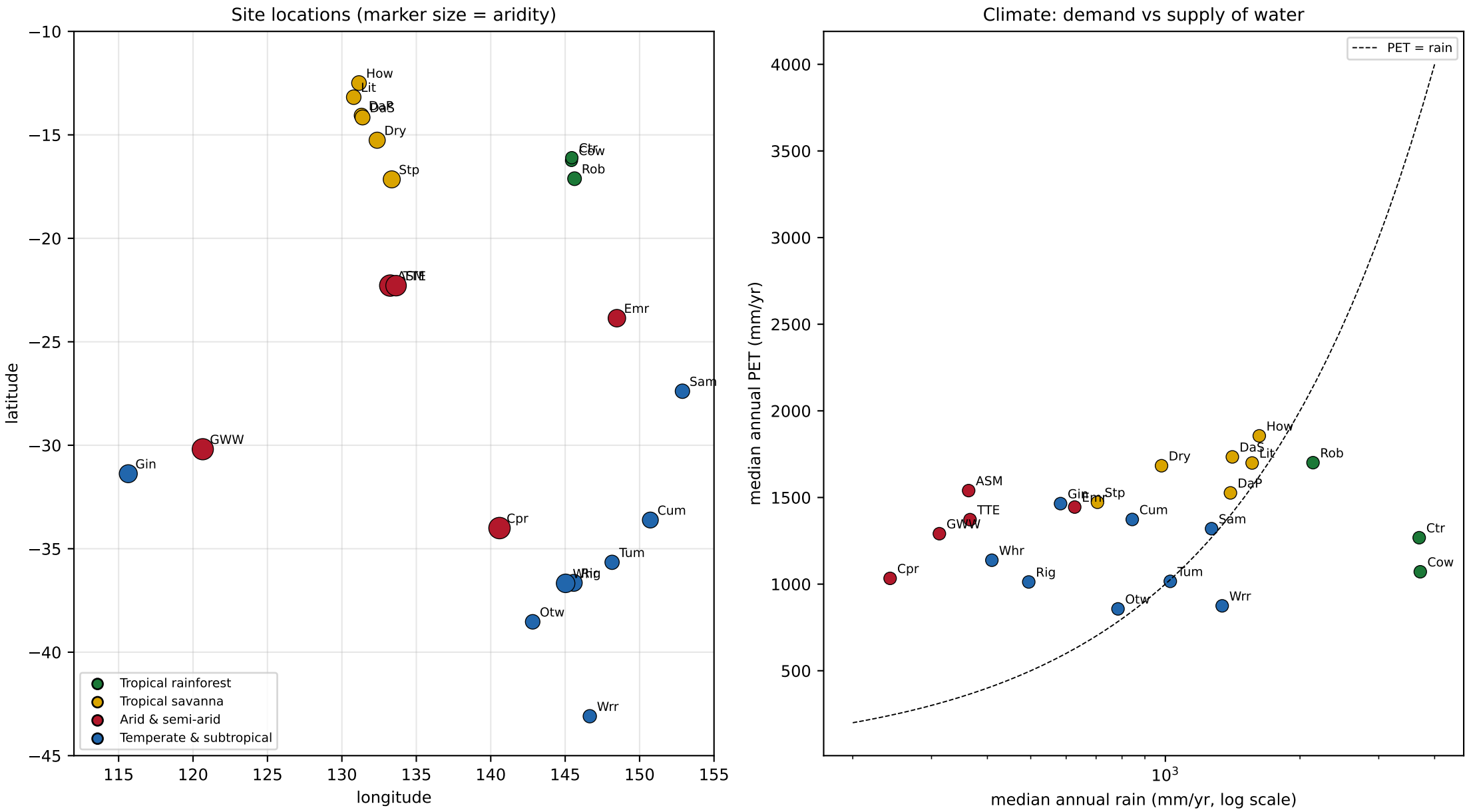
Site groups. To read patterns, sites are grouped into tropical rainforest (AU-Cow, AU-Ctr, AU-Rob), tropical savanna (AU-How, AU-Lit, AU-DaS, AU-Dry, AU-DaP, AU-Stp), arid & semi-arid (AU-ASM, AU-TTE, AU-GWW, AU-Cpr, AU-Emr) and temperate & subtropical (AU-Tum, AU-Wrr, AU-Whr, AU-Rig, AU-Otw, AU-Cum, AU-Sam, AU-Gin). The aridity index is PET / rain: below 1 is wet (energy-limited), above 2 is dry (water-limited).

Table 1. The 22 sites, ordered from wettest to driest (aridity = PET / rain). Grey rows: fewer than 3 complete years.

Site	Name	Veg	Group	Lat	Rain mm/yr	PET mm/yr	Aridity PET/P	LAI	Years	Rain events	Energy gap %	Tower ET mm/yr
AU-Cow	Cow Bay	EBF	Rainforest	-16.2	3715	1071	0.29	6.2	6	482	15	1182
AU-Ctr	Cape Tribulation	EBF	Rainforest	-16.1	3694	1268	0.34	5.1	8	718	15	1390
AU-Wrr	Warra	EBF	Temperate	-43.1	1340	875	0.65	2.5	2	86	42	586
AU-Rob	Robson Creek	EBF	Rainforest	-17.1	2139	1701	0.80	6.1	4	354	37	1154
AU-Tum	Tumbarumba	EBF	Temperate	-35.7	1026	1016	0.99	3.5	16	533	28	900
AU-Sam	Samford	GRA	Temperate	-27.4	1269	1320	1.04	1.9	7	210	37	631
AU-Lit	Litchfield	WSA	Savanna	-13.2	1563	1699	1.09	1.5	2	88	13	1313
AU-DaP	Daly Pasture	GRA	Savanna	-14.1	1398	1526	1.09	1.6	4	161	24	751
AU-Otw	Otway	GRA	Temperate	-38.5	784	857	1.09	3.1	2	44	21	674
AU-How	Howard Springs	WSA	Savanna	-12.5	1622	1856	1.14	1.8	15	734	19	1204
AU-DaS	Daly Uncleared	SAV	Savanna	-14.2	1412	1734	1.23	1.4	8	301	8	948
AU-Cum	CUP_EC	EBF	Temperate	-33.6	843	1373	1.63	1.4	6	137	43	805
AU-Dry	Dry River	SAV	Savanna	-15.3	980	1683	1.72	1.1	5	139	29	821
AU-Rig	Riggs Creek	GRA	Temperate	-36.6	495	1012	2.05	1.5	6	90	39	470
AU-Stp	Sturt Plains	GRA	Savanna	-17.2	705	1472	2.09	0.6	8	184	-2	560
AU-Emr	Arcturus	GRA	Arid	-23.9	627	1445	2.30	0.9	2	38	65	410
AU-Gin	Gingin	WSA	Temperate	-31.4	583	1465	2.51	1.0	6	117	15	573
AU-Whr	Whroo	EBF	Temperate	-36.7	409	1138	2.78	1.1	2	19	22	510
AU-TTE	TiTreeEast	OSH	Arid	-22.3	366	1372	3.75	0.3	5	60	-1	337
AU-GWW	Great Western Wo	SAV	Arid	-30.2	312	1291	4.13	0.5	5	35	28	259
AU-ASM	AliceSpringsMulg	ENF	Arid	-22.3	363	1540	4.24	0.4	7	81	29	286
AU-Cpr	Calperum	SAV	Arid	-34.0	242	1033	4.26	0.5	7	49	-7	265

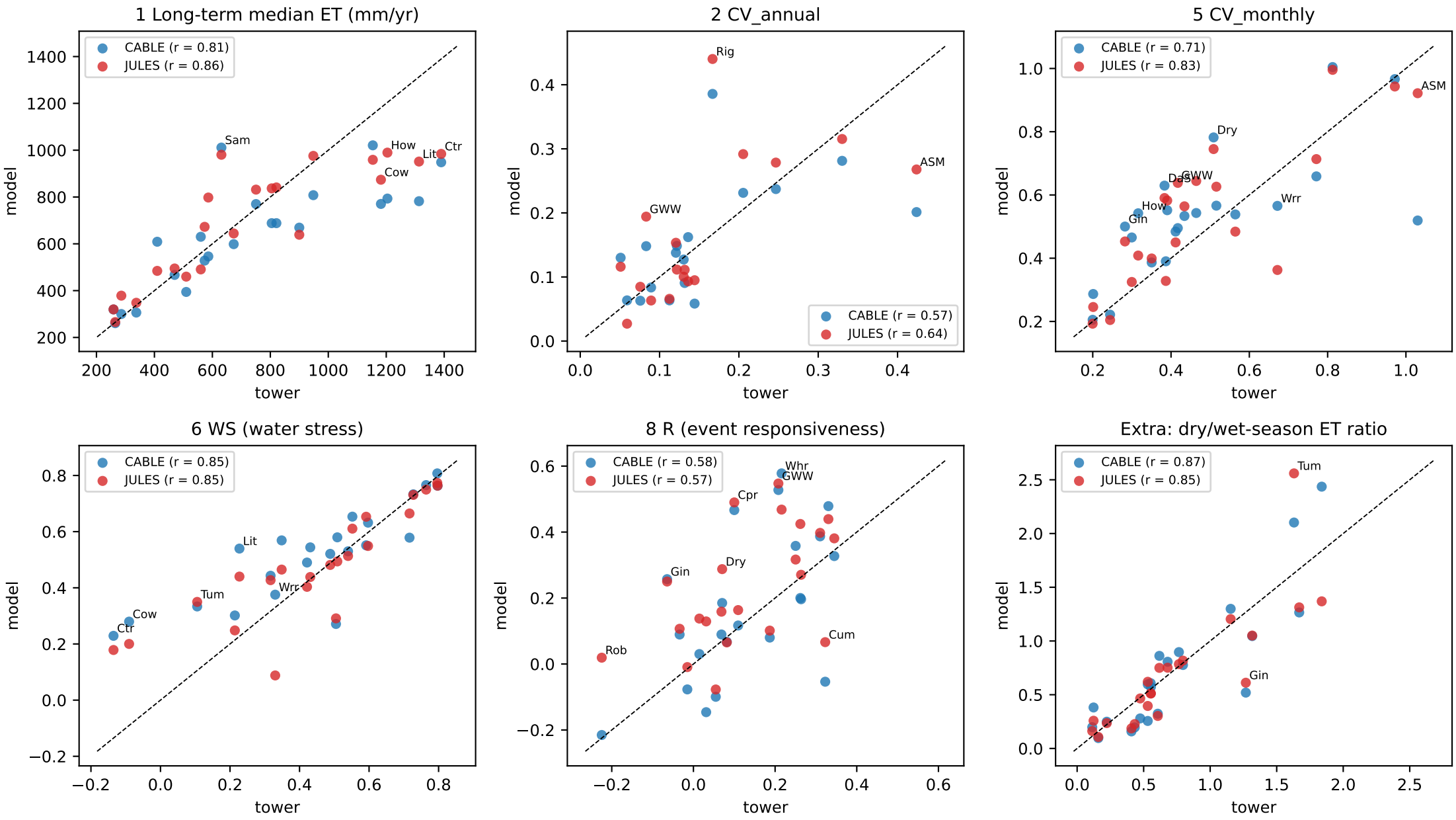
Energy gap % = how much more energy the tower's energy-balance-corrected fluxes contain than the raw ones (Qle_cor + Qh_cor vs Qle + Qh, measured steps): a larger gap means the tower misses more energy. LAI = mean satellite leaf area index from the tower file. Tower ET = metric 1 (gap-filled tower).

Figure 1. Where the sites are and what climates they span



The sites cover monsoon tropics (north), wet tropical rainforest (north-east), the arid interior and temperate south. Sites left of the dashed line receive less rain than the atmosphere could evaporate (water-limited); the rainforests on the right are energy-limited. There is no coastline: positions are plotted as raw latitude/longitude.

Figure 2. Model vs tower for each signature (one point per site; dashed = 1:1)



Points on the dashed line are perfect. Labelled sites are the largest misses. Annual ET and WS are well ordered across sites; R and CV_annual much less so. CV_annual uses only the 17 sites with at least 3 complete years.

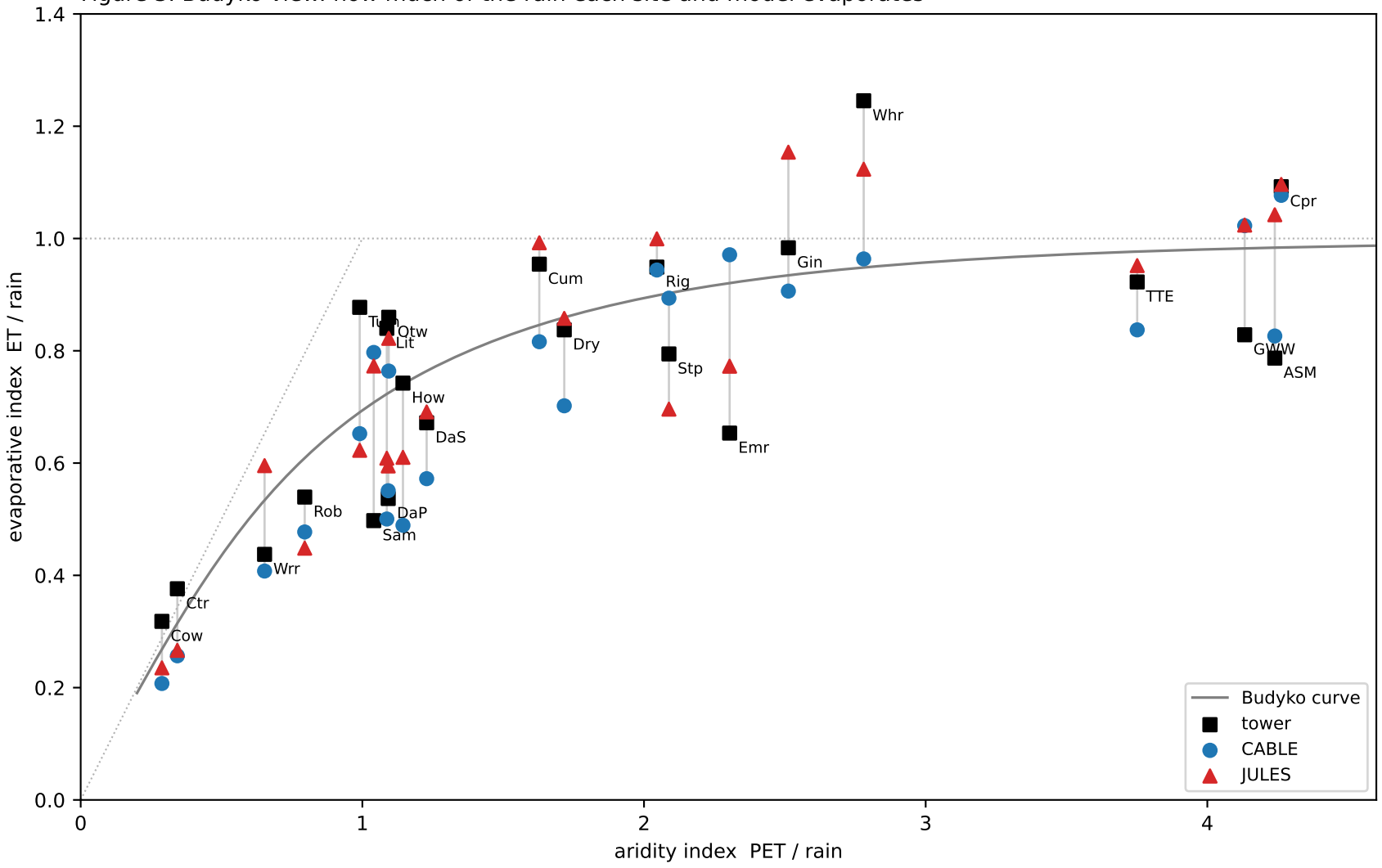
Table 2a. How well each model reproduces each signature across the 22 sites (error = model - tower)

Model	Metric	n	r across sites	median error	mean error	RMSE
CABLE	1 Long-term median ET (mm/yr)	22	0.81	-60	-96.2	229
CABLE	2 CV_annual	17	0.57	-0.00315	-0.000866	0.0856
CABLE	5 CV_monthly	22	0.71	+0.0753	+0.0549	0.172
CABLE	6 WS (water stress)	22	0.85	+0.057	+0.079	0.166
CABLE	8 R (event responsiveness)	22	0.58	+0.0125	+0.0434	0.186
CABLE	Extra: dry/wet-season ET ratio	22	0.87	+0.0218	-0.0237	0.293
JULES	1 Long-term median ET (mm/yr)	22	0.86	+15.3	-36.7	185
JULES	2 CV_annual	17	0.64	-0.0148	+0.0107	0.0891
JULES	5 CV_monthly	22	0.83	+0.0466	+0.0541	0.142
JULES	6 WS (water stress)	22	0.85	-0.00166	+0.0346	0.14
JULES	8 R (event responsiveness)	22	0.57	+0.094	+0.102	0.183
JULES	Extra: dry/wet-season ET ratio	22	0.85	-0.0248	-0.0568	0.3

Table 2b. Head to head: at how many sites is each model closer to the tower?
Green = significant at $p < 0.05$. For TSP the counts are sites with the exact peak month.

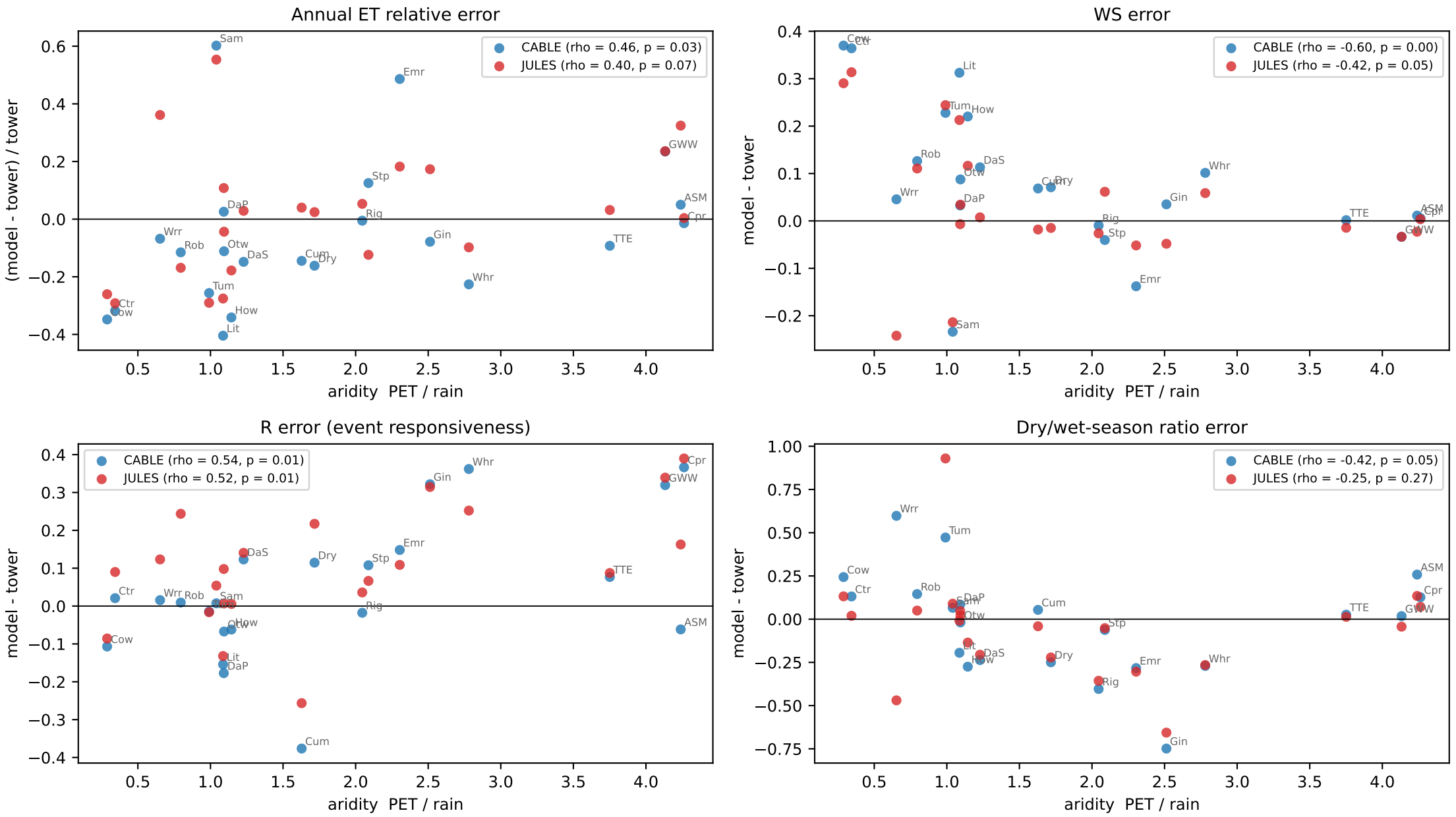
Metric	n	sites JULES closer	sites CABLE closer	median |err| CABLE	median |err| JULES	Wilcoxon p
1 Long-term median ET (mm/yr)	22	14	8	116	77.8	0.137
2 CV_annual	17	8	9	0.0268	0.0328	0.487
5 CV_monthly	22	10	12	0.0925	0.1	0.975
6 WS (water stress)	22	15	7	0.0793	0.0499	0.028
8 R (event responsiveness)	22	10	12	0.107	0.116	0.899
Extra: dry/wet-season ET ratio	22	17	5	0.216	0.111	0.006
4 TSP (exact peak month)	22	14	9	1.0 months	0.0 months	-

Figure 3. Budyko view: how much of the rain each site and model evaporates



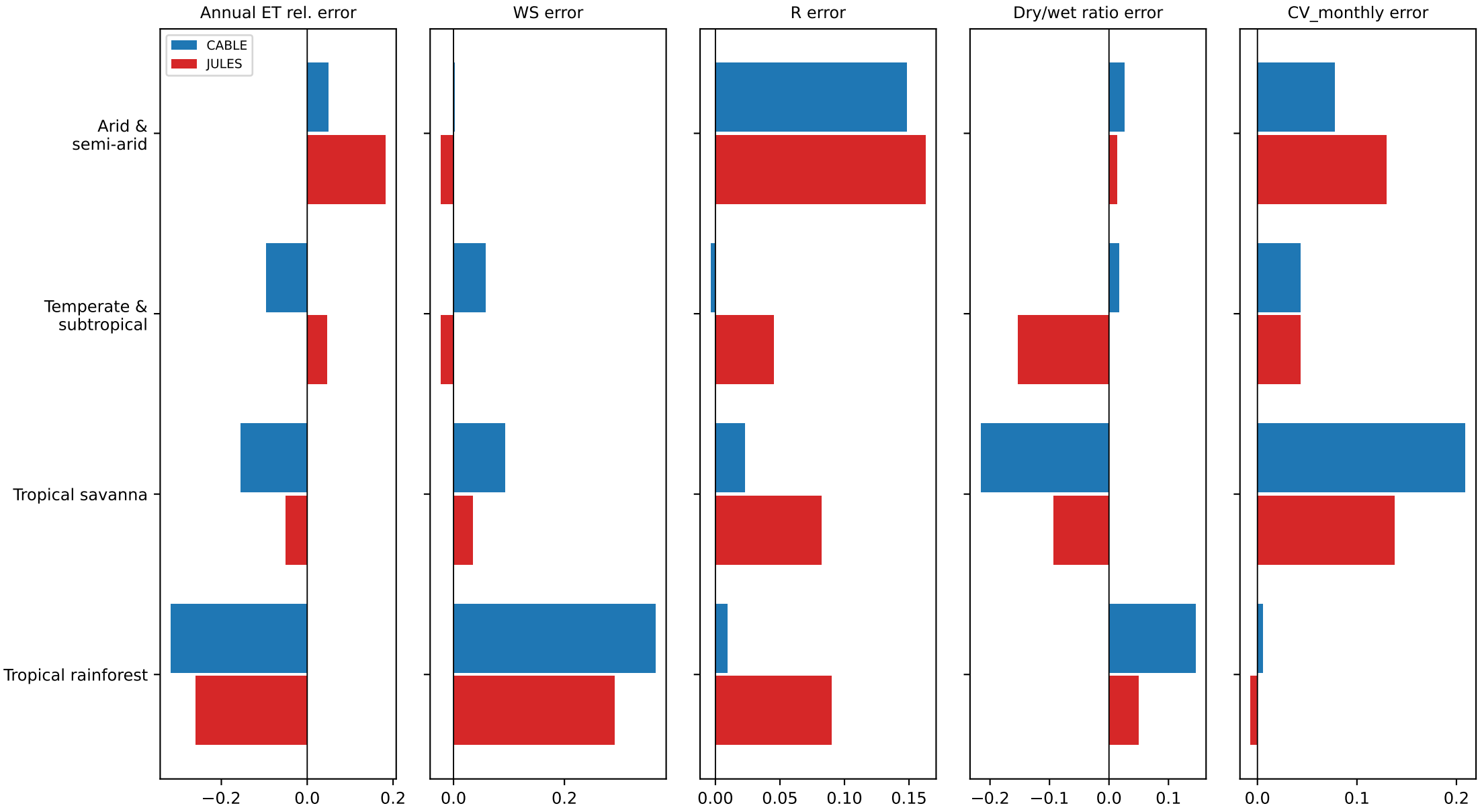
Each site has one x position (its climate) and three points (tower, CABLE, JULES), joined by a grey line. The Budyko curve is the classic expectation for catchments. Wet sites (left) are under-evaporated by both models, most at the rainforests and AU-How/AU-Lit. Dry sites (right) sit near ET/rain = 1 for everyone, because rain is the cap. Points above 1 (e.g. AU-Whr tower, JULES at several arid sites) evaporate more than it rains, which needs stored or extra water.

Figure 4. Do the errors depend on how dry the site is?



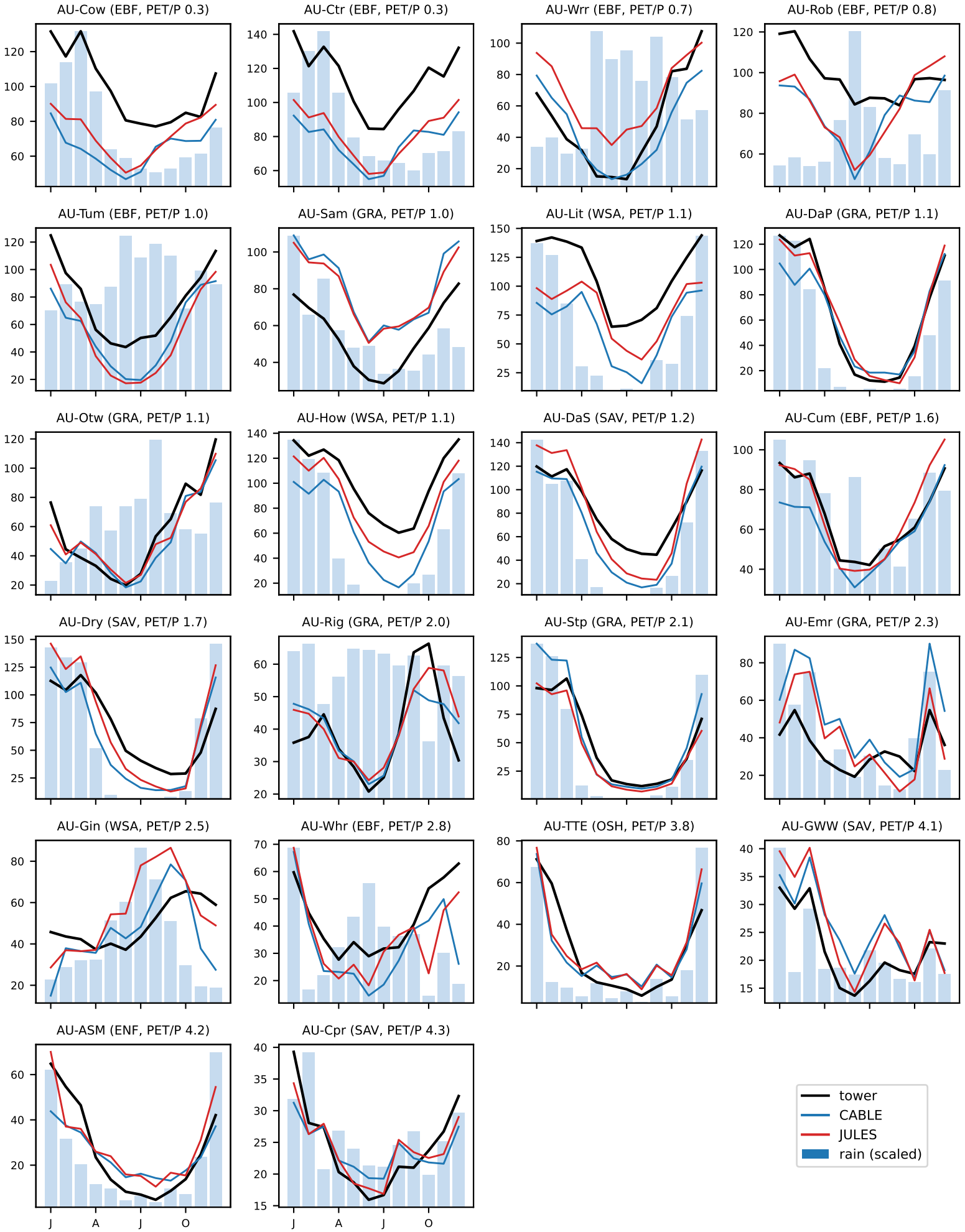
rho = Spearman rank correlation of error with aridity across the 22 sites. Labels are placed at CABLE's point. Annual ET goes from under-estimated at wet sites to slightly over-estimated at dry sites (significant for CABLE). Excess rain responsiveness (R) grows clearly with dryness for both models.

Figure 5. Median error by site group (model - tower)



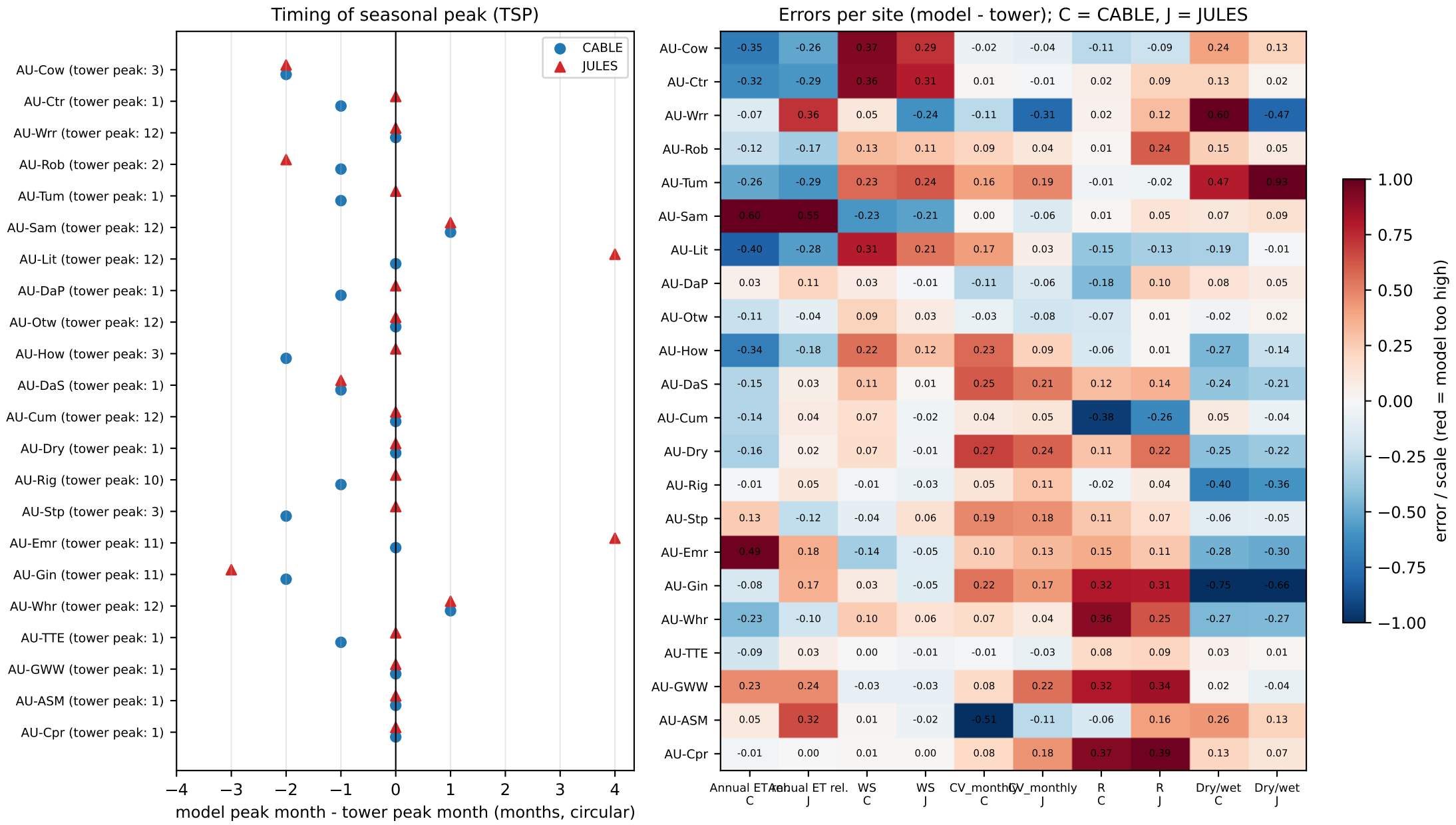
Annual ET relative error is a fraction (-0.3 = 30% too little). The rainforests stand out for annual ET and WS; the savannas for the dry/wet ratio and CV_monthly (CABLE especially); the arid sites for R. Groups have 3-8 sites, so these are indicative.

Figure 6. Mean seasonal cycle of ET (mm/month), wettest site first



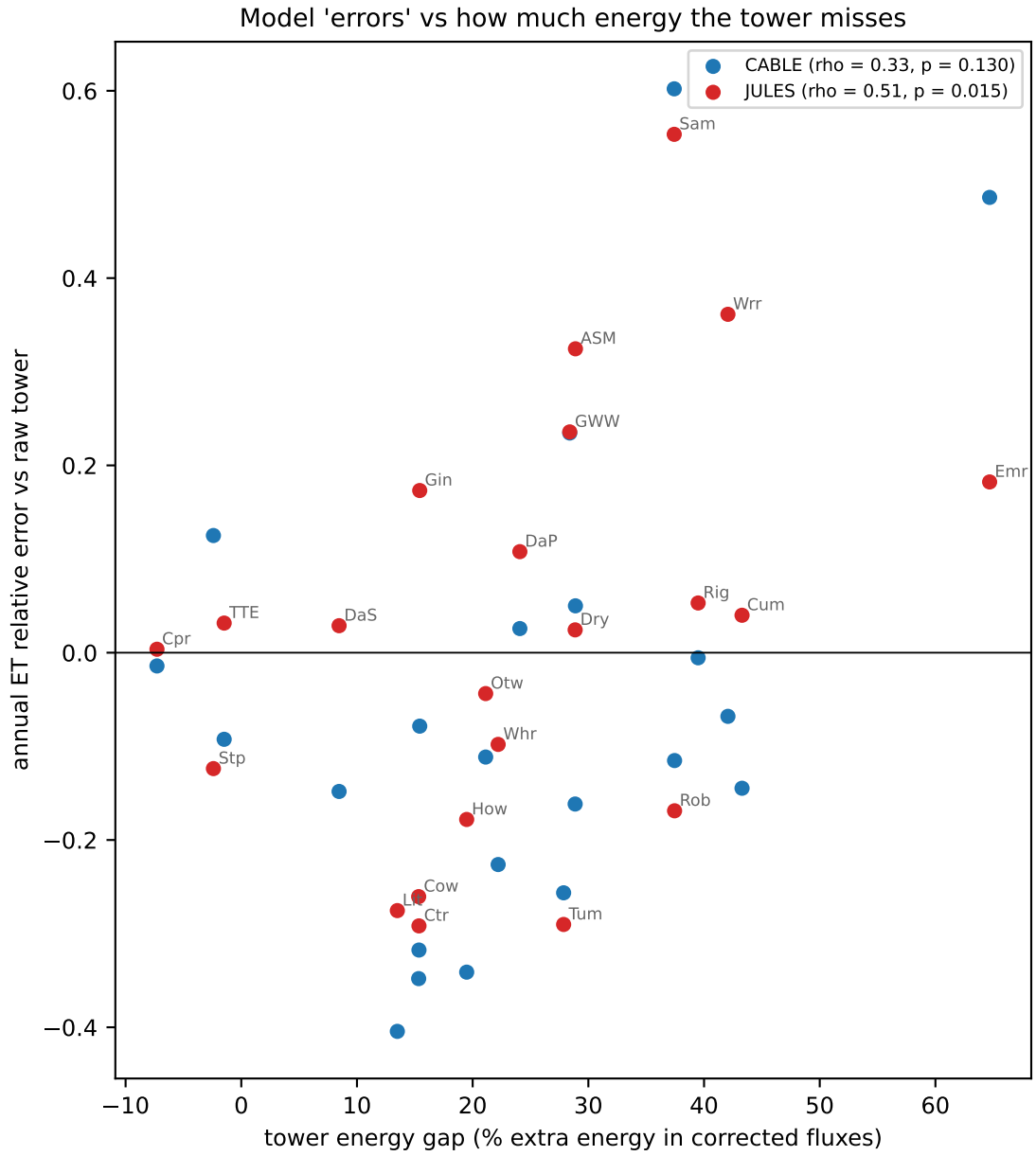
Blue bars: mean monthly rain (own scale). Note the dry-season gap between the tower and CABLE at the northern savannas (How, DaS, Dry, Lit), the dry-season peak at Gin that neither model has, and the rainforest under-estimate in every month.

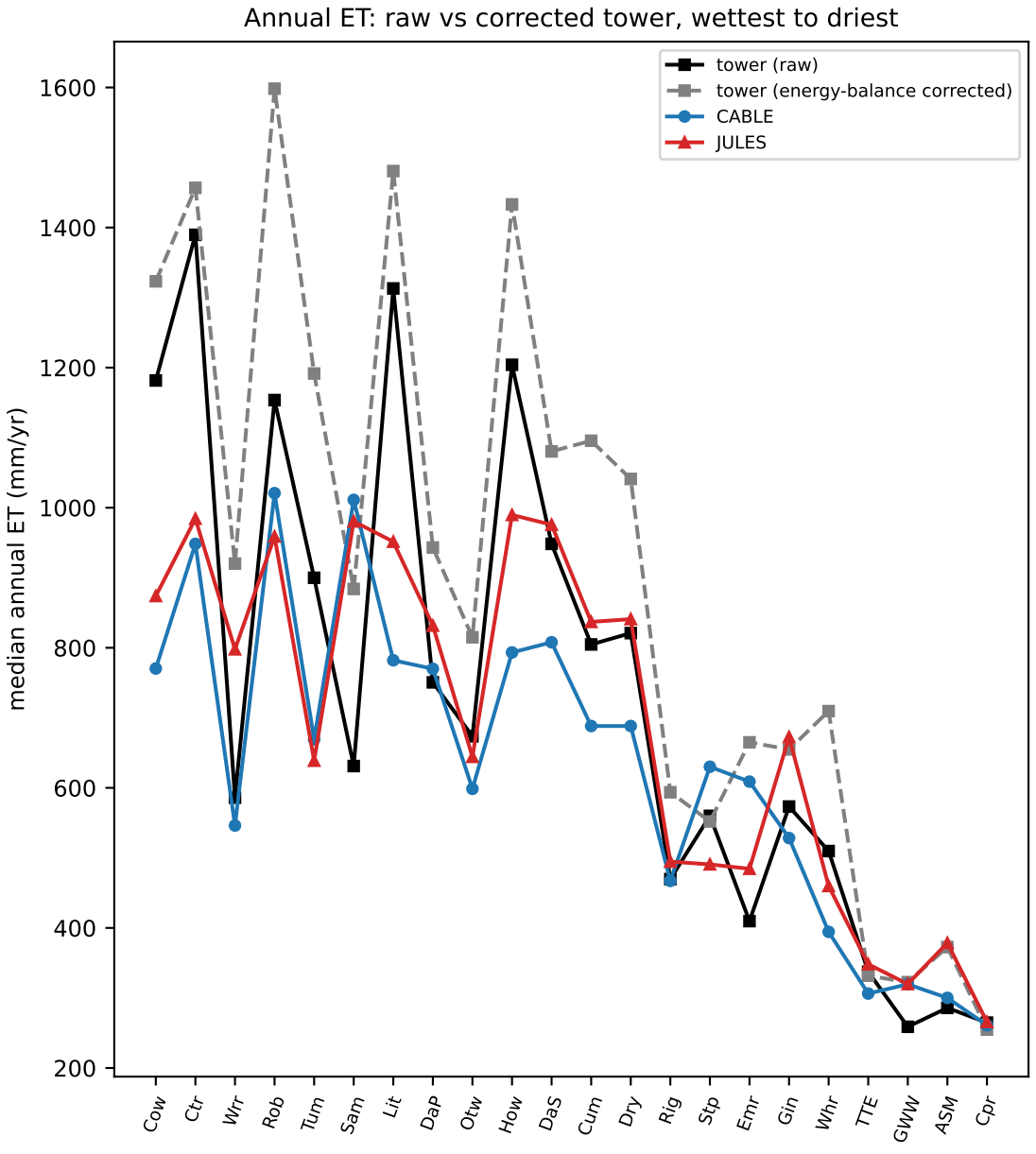
Figure 7. Site-by-site detail, wettest site at the top



Left: CABLE's seasonal peak is early or on time almost everywhere (it is never more than one month late); JULES is exactly right at most sites but far off at a few short or unusual records (Emr, Lit, Gin). Right: numbers are the errors; colours are scaled per column so each metric is readable. At the top (rainforests), annual ET is blue (too little) and WS is red (too much water stress).

Figure 8. Part of the 'model error' is tower error





Towers typically measure less energy than the radiation budget says is available. Where that gap is large (Emr, Cum, Wrr, Rig, Rob, Sam) the raw tower ET is probably too low, and JULES's apparent over-estimates there shrink or reverse against the corrected ET. Against corrected ET both models under-estimate at most wet and mid sites.

2. Findings in detail

2.1 Annual totals (metric 1)

Both models order the sites correctly: annual ET correlates with the tower at r = 0.81 (CABLE) and 0.86 (JULES). This is largely because climate sets most of the between-site differences, and both models are forced with the real weather. The errors are not random, though. Relative error rises with aridity (CABLE rho = 0.46, p = 0.03; JULES rho = 0.40, p = 0.07) and falls with leaf area (CABLE rho = -0.49, JULES rho = -0.45): the leafiest, wettest sites are under-evaporated, the dry sparse sites roughly right or slightly over. The worst cases are the lowland tropical rainforests AU-Cow and AU-Ctr (-26% to -35%), AU-Lit (CABLE -40%), AU-How (CABLE -34%), AU-Tum (both -26% to -29%) and, in the other direction, AU-Sam (both +55% to +60%) and AU-Emr (CABLE +49%, JULES +18%).

2.2 Shared errors

CABLE's and JULES's annual-ET errors are strongly correlated across sites (rho = 0.80). At 8 sites both under-estimate and at 5 both over-estimate. Two very different models failing the same way at the same sites suggests a common cause. Candidates: (a) a process both lack, such as deep roots or groundwater; (b) PLUMBER2's out-of-the-box vegetation and soil parameters; (c) problems in the tower data itself (Section 2.6). Shared errors are exactly the 'many models fail' signal Gab described as most valuable.

2.3 Seasonality (metrics 4 and 5, and the dry/wet ratio)

Peak timing is mostly right: JULES matches the tower's peak month at 14 sites and CABLE at 9; within one month, 17 and 18. CABLE's misses are almost all early (mean offset -0.6 months): it evaporates hardest early in the wet season and fades before the tower does. The seasonal swing (CV_monthly) is too big in both models (median +0.08 CABLE, +0.05 JULES), and for CABLE that excess is tied to the dry season: sites where CABLE's dry/wet ratio is too low are those where its CV_monthly is too high (rho = -0.58). In other words the exaggerated swing is a dry-season problem, not a wet-season one.

2.4 Dry-season persistence (extra diagnostic)

The tropical savannas show the clearest model failure in this analysis. In the dry season the tower at AU-How still evaporates about half of its wet-season rate (ratio 0.53); CABLE drops to 0.26 and JULES to 0.40. AU-DaS (0.43 vs 0.20/0.23) and AU-Dry (0.41 vs 0.16/0.19) are similar. Savanna trees reach deep soil water through the dry season; a model with too-shallow roots or too-strict water-stress response runs dry. JULES is significantly closer on this ratio (17 of 22 sites, p = 0.006). The same signature appears at AU-Gin, more extremely: the tower's dry/wet ratio is 1.27 (more ET in the dry summer), against 0.52 and 0.61 for the models. Gingin sits on deep sand above a shallow aquifer, and the vegetation is known to use groundwater; neither model can.

2.5 Water stress and event responsiveness (metrics 6 and 8)

Because the tower and both models share the same PET, WS is almost the mirror image of the annual-ET error (rho = -0.96 for CABLE, -0.94 for JULES), so it adds little new information here. It does expose the PET substitute: at AU-Cow and AU-Ctr the tower evaporates more than our Priestley-Taylor PET (negative WS), which is physically possible for wet canopies and warm advected air but suggests PET is low there. R is the most interesting dynamic signature. At wet sites R is near zero for everyone. At dry sites the tower's R stays modest (0.10-0.34), but both models' R climbs to 0.4-0.55: modelled ET jumps after every big storm and then falls, while the real ecosystem buffers rain in deeper soil and releases it slowly. The excess grows with aridity (CABLE rho = 0.54, p = 0.010; JULES rho = 0.52, p = 0.013).

2.6 The tower data are part of the answer

The raw tower fluxes miss between -7% and +65% of the available energy (Table 1). JULES's relative error correlates with that gap (rho = 0.51, p = 0.015; CABLE rho = 0.33): the sites where JULES seems to over-evaporate (AU-Sam, AU-Emr, AU-Wrr, AU-Cum) are among those with the largest gaps, where the raw tower ET is probably too low. Against energy-balance-corrected ET, both models under-evaporate (median -26% CABLE, -18% JULES). The paper itself uses the raw (uncorrected) OzFlux ET, so our main results follow it, but any claim about the sign of a model bias should be checked against both versions.

Table 3. Site by site, wettest to driest

Site	Group	What stands out
AU-Cow	Tropical rainforest	Wet rainforest. Both models 26-35% too little ET in every month; tower ET above PT PET.
AU-Ctr	Tropical rainforest	Wet rainforest. Largest absolute miss (-440 mm/yr CABLE). CABLE peaks a month early.
AU-Wrr	Temperate/subtropical	Tall wet forest, 2 years only. JULES +36% ET, CABLE about right; WS very low in JULES.
AU-Rob	Tropical rainforest	Upland rainforest, 37% energy gap. Both ~12-17% low; peaks 1-2 months early.
AU-Tum	Temperate/subtropical	Tall temperate forest. Both 26-29% too little ET, worse against corrected ET; summer peak too sharp.
AU-Sam	Temperate/subtropical	Subtropical pasture. Both +55-60% ET; tower ET only half the rain and 37% energy gap: suspect tower.
AU-Lit	Tropical savanna	Savanna, 2 years. CABLE -40% ET; JULES peak 4 months late (short record).
AU-DaP	Tropical savanna	Monsoon pasture. Totals within 3-11%; dry-season ET a little too high in both models.
AU-Otw	Temperate/subtropical	Temperate pasture, 2 years. Close on everything; a good 'control' site.
AU-How	Tropical savanna	Savanna, 15 years. CABLE -34%, JULES -18%; dry-season collapse (0.26/0.40 vs 0.53). Key case.
AU-DaS	Tropical savanna	Savanna. JULES total right; both collapse in dry season (0.20/0.23 vs 0.43).
AU-Cum	Temperate/subtropical	Dry sclerophyll forest, 43% energy gap. CABLE -14%, JULES +4%; R far too low in CABLE.
AU-Dry	Tropical savanna	Savanna. Dry-season collapse (0.16/0.19 vs 0.41); seasonal swing far too big.
AU-Rig	Temperate/subtropical	Temperate grassland. Totals right; interannual variability 2-3x too high in both.
AU-Stp	Tropical savanna	Tussock grassland. Totals within ~12%; dry-season ET too low; swing too big.
AU-Emr	Arid/semi-arid	Semi-arid grassland, 2 years, 65% energy gap. CABLE +49%; JULES peak 4 months off.
AU-Gin	Temperate/subtropical	Banksia woodland on sand. Tower peaks in dry summer (groundwater); neither model can.
AU-Whr	Temperate/subtropical	Box woodland, 2 years. Tower ET above rain (1.25); models R too high.
AU-TTE	Arid/semi-arid	Arid shrubland. Very close on totals, seasonality and WS; R slightly high.
AU-GWW	Arid/semi-arid	Semi-arid woodland. Both +24% ET and R ~0.54 vs 0.21: too responsive to rain.
AU-ASM	Arid/semi-arid	Mulga. Totals close (JULES +32%); interannual variability too low; JULES R high.
AU-Cpr	Arid/semi-arid	Mallee. Totals exact; R 0.47/0.49 vs 0.10: ET reacts to every storm in the models.

3. Caveats

- PET is a substitute (Priestley-Taylor from tower net radiation) for the paper's Morton PET. It is likely too low at the wet tropical sites (tower ET exceeds it). WS and the aridity ordering depend on it; the other metrics do not.
- Tower ET is gap-filled (as in the paper), and 11-51% of half-hours at each site are gap-filled. The raw tower data also miss energy (Section 2.6).
- Five sites have only 2 complete years (AU-Emr, AU-Lit, AU-Otw, AU-Whr, AU-Wrr). Their CV_annual is excluded and their other metrics are noisy.
- R uses a 10 mm/day rain threshold; the paper does not state its value. Events less than 10 days apart share their windows.
- 22 sites is a small sample. Wilcoxon and Spearman p-values are indicative; with 6 metrics tested, one 'significant' result could be chance.
- The model runs are PLUMBER2 'out of the box' runs. Which CABLE and JULES configurations these are (CABLE vs CABLE-POP, JULES GL9 vs GL9_LAI) is still to be confirmed.
- AU-Ync is excluded: its tower files contain AU-ASM data. Note that et_signatures.py currently loops over every file in data/harmonised, including AU-Ync.

4. What this suggests doing next

- Get Morton PET from SILO for each site and rerun, to remove the PET caveat and match the paper.
- Rerun every metric against energy-balance-corrected tower ET and report both; it changes the sign of JULES's bias at several sites.
- Use the dry-season result as a case study: the AU-How, AU-DaS and AU-Dry dry seasons, with soil moisture and transpiration from the CABLE and JULES outputs, to test whether root depth or the water-stress function is the cause.
- AU-Gin as a groundwater test case, and the arid sites (AU-Cpr, AU-GWW) for the too-responsive-to-rain behaviour: how fast does modelled ET fall after a storm compared with the tower (dry-down time scale)?
- Check AU-Sam and AU-Emr tower data quality before treating them as model failures.
- Repeat the signatures for the other 11 PLUMBER2 models to see which errors are shared by many models (a general process gap) and which are specific to CABLE or JULES.